

Incomplete Information in Game Theory

Incomplete-information games model strategic situations in which at least one player does not know some payoff-relevant fact, such as another player's cost, valuation, capability, or objective. This guide develops the five ideas most often tested in interviews: **types of players, Bayesian games, prior probabilities, expected payoff, and Bayesian Nash equilibrium.**

1. Overview

Definition

A game has **incomplete information** when a player does not know the complete structure relevant to decisions—usually another player's payoff function or private characteristic. The hidden characteristic is represented by a player's **type**.

A finite Bayesian game is commonly written as:

$$G = (N, (A_i), (T_i), p, (u_i))$$

where:

- N is the set of players.
- A_i is player i 's action set.
- T_i is player i 's type set.
- $p(t_1, \dots, t_n)$ is the common prior over type profiles.
- $u_i(a_1, \dots, a_n, t_1, \dots, t_n)$ is player i 's payoff.

A **Bayesian Nash equilibrium (BNE)** is a strategy profile in which every type of every player maximizes its expected payoff, given its beliefs and the other players' strategies.

Why it matters

Most real strategic systems contain private information. Buyers know their valuations, cloud tenants know their urgency, services know their local load, and attackers know their capabilities. A complete-information model can predict the wrong behavior if it ignores these differences.

Where it is used in real systems

- **Online auctions:** bidders privately know how much an item or ad impression is worth.
- **Cloud scheduling:** jobs privately know urgency, resource demand, or willingness to pay.
- **Networking:** endpoints have private traffic demand or quality requirements.
- **Cybersecurity:** defenders are uncertain about attacker type, intent, and capability.
- **Distributed systems:** nodes may be honest, faulty, selfish, or malicious.
- **Marketplaces:** sellers privately know quality or production cost.
- **Hiring and signaling:** candidates know their ability more accurately than employers.

Why interviewers ask about it

The topic tests whether you can:

1. Distinguish an unknown action from an unknown type.
2. formalize beliefs as probabilities rather than vague uncertainty.
3. calculate conditional expected utility correctly.
4. define a strategy as a mapping from type to action.
5. verify equilibrium separately for every possible type.

2. Core Idea

Intuition

In a complete-information game, a player asks, "What will the other player do?" In an incomplete-information game, the player must also ask, "What kind of player am I facing?"

Nature conceptually chooses the hidden types before strategic play:

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Nature draws type profile  $t$  from prior  $p$ 
      |
      +--> Player  $i$  observes its own type  $t_i$ 
      +--> Player  $i$  may not observe opponents' types  $t_{-i}$ 
      |
      v
Players choose actions using strategies  $s_i(t_i)$ 
      |
      v
Payoffs depend on actions and types
  
```

The "Nature" step is a modeling device. Nature does not optimize; it represents chance.

Real-world analogy

Imagine bidding in a sealed-bid auction. You know your valuation is ₹80,000, but you do not know the other bidder's valuation. You have a belief about it—perhaps it is uniformly distributed between ₹0 and ₹100,000. Your bid should balance two effects:

- A higher bid increases the probability of winning.
- A higher bid reduces your surplus if you win.

Your best action therefore depends on your private type and on beliefs about the other bidder's type.

Small example: deploy or wait

Two teams may deploy to a shared production service. Team A knows whether its release is **urgent** (U) or **routine** (R). Team B sees only the prior:

$$P(A \text{ is urgent}) = 0.3$$

$$P(A \text{ is routine}) = 0.7$$

Suppose B chooses **Deploy** or **Wait**. Its payoff is:

A's type and likely action	B deploys	B waits
Urgent A deploys	-6	1
Routine A waits	4	2

If B believes urgent A deploys and routine A waits:

$$EU_B(\text{Deploy}) = 0.3(-6) + 0.7(4) = 1.0$$

$$EU_B(\text{Wait}) = 0.3(1) + 0.7(2) = 1.7$$

So B waits. Notice that B does not maximize against one known row; it maximizes an expectation over possible types.

Step-by-step explanation

1. **Identify private facts.** A's release urgency is private information.
2. **Create types.** $T_A = \{U, R\}$.
3. **Assign a prior.** B initially believes $P(U)=0.3$ and $P(R)=0.7$.
4. **Define actions.** Both teams can deploy or wait.
5. **Define type-dependent payoffs.** Collision is especially costly when both deploy.
6. **Define strategies by type.** A strategy for A must say what urgent A does and what routine A does.
7. **Calculate expected payoffs.** Weight each state by its probability.
8. **Check incentives.** A BNE requires B's action and both of A's type-contingent actions to be optimal.

3. Important Subtopics

3.1 Types of players

What it means

A **type** packages everything privately known by a player that affects preferences, feasible actions, or relevant beliefs. Types can describe:

- high or low valuation;
- low or high production cost;
- urgent or delay-tolerant workload;
- honest, selfish, faulty, or malicious behavior;
- strong or weak capability;
- risk tolerance or private signal.

The phrase “types of players” does not merely classify people informally. In a Bayesian model, t_i is a formal variable and a pure strategy is a function:

$$s_i : T_i \rightarrow A_i$$

If $T_i = \{\text{High}, \text{Low}\}$ and $A_i = \{\text{Aggressive}, \text{Conservative}\}$, a pure strategy must specify two actions, such as:

```
s_i(High) = Aggressive
s_i(Low)  = Conservative
```

Why it matters

Two types of the same player can rationally choose different actions because their payoffs differ. Checking only one type is not enough to establish equilibrium.

Example

In a cloud auction, a batch job values immediate execution at ₹20 while a latency-sensitive job values it at ₹100. The job's type changes its willingness to bid.

Common interview angle

Interviewers ask, “What belongs in a type?” A good answer is: all payoff-relevant private information needed to make the game complete once types are fixed. Avoid putting observable actions into the type unless they genuinely represent information selected before play.

3.2 Bayesian games

What it means

A **Bayesian game** combines actions, private types, beliefs, and type-dependent payoffs. Each player observes its own type and has beliefs about unknown types. In the standard Harsanyi model, a common prior generates all types.

Why it matters

It turns uncertainty about the game itself into uncertainty inside a larger, well-defined game. Conditional on the realized type profile, payoffs are fully specified.

Example

In intrusion response, the defender chooses **Monitor** or **Block**; the attacker chooses **Attack** or **Leave**. The attacker may be **Skilled** or **Unskilled**. Blocking can be necessary against a skilled attacker but unnecessarily expensive against an unskilled one.

Common interview angle

Be able to list the tuple and explain the timing: Nature draws types, players observe specified information, then choose actions. Also distinguish a Bayesian game from a stochastic game, where a public or private state evolves over multiple periods.

3.3 Prior probabilities and posterior beliefs

What it means

A **prior** is a probability distribution over types before new strategic evidence is observed. A **posterior** is the updated belief after observing a signal or action.

Bayes' rule is:

$$P(t \mid x) = P(x \mid t)P(t) / P(x)$$

where x is observed evidence.

Why it matters

Expected payoffs and best responses depend on beliefs. Incorrect conditioning is one of the most common errors in Bayesian-game calculations.

Example

Suppose 20% of nodes are faulty. A timeout occurs with probability 0.8 for faulty nodes and 0.1 for healthy nodes:

$$\begin{aligned} P(\text{Faulty} \mid \text{Timeout}) &= (0.8)(0.2) / [(0.8)(0.2) + (0.1)(0.8)] \\ &= 0.16 / 0.24 \\ &= 2/3 \end{aligned}$$

The timeout raises the failure belief from 0.2 to about 0.667.

Common interview angle

State the conditioning event explicitly. A player's relevant belief is usually $P(t_{-i} \mid t_i)$ rather than an unconditional marginal. If types are correlated, observing one's own type can reveal information about others.

3.4 Expected payoff

What it means

Expected payoff is probability-weighted utility. For player i , type t_i , candidate action a_i , and opponents' strategies s_{-i} :

$$\begin{aligned} & EU_i(a_i \mid t_i, s_{-i}) \\ &= \text{sum over } t_{-i} \text{ of } P(t_{-i} \mid t_i) \\ &\quad * u_i(a_i, s_{-i}(t_{-i}), t_i, t_{-i}) \end{aligned}$$

If opponents mix, also average over their randomized actions.

Why it matters

Players cannot optimize against an unknown state directly. Under expected-utility assumptions, they compare probability-weighted outcomes.

Example

If a security action gives payoff 8 against a weak attacker and -4 against a strong attacker, with probabilities 0.75 and 0.25:

$$EU = 0.75(8) + 0.25(-4) = 5$$

Common interview angle

Do not average actions or types first and then plug the “average type” into a nonlinear payoff. Evaluate payoff state by state, multiply by probability, and sum.

3.5 Bayesian Nash equilibrium

What it means

A strategy profile s^* is a BNE if, for every player i and every type t_i with positive probability:

$$\begin{aligned} & s_i^*(t_i) \text{ belongs to argmax over } a_i \text{ of} \\ & \text{sum over } t_{-i} P(t_{-i} \mid t_i) \\ & u_i(a_i, s_{-i}^*(t_{-i}), t_i, t_{-i}) \end{aligned}$$

No type can improve its expected payoff through a unilateral deviation.

Why it matters

BNE is the basic stability concept for simultaneous games with private information. It underlies auctions, mechanism design, signaling analysis, and resource-allocation protocols.

Example: threshold decision

A server accepts a job if its private value v exceeds expected congestion cost c . If accepting yields $v - c$ and rejecting yields 0, its best response is:

```
Accept if  $v \geq c$   
Reject if  $v < c$ 
```

This is a type-dependent threshold strategy, a common form of BNE.

Common interview angle

The equilibrium object is a **profile of functions**, not merely an action pair. Verify incentive compatibility for each type, including types that occur with small but positive probability.

3.6 Harsanyi transformation

What it means

The Harsanyi transformation represents incomplete information by adding an initial chance move that selects types. Players then know whatever the information structure says they know.

Why it matters

It allows ordinary strategic reasoning over an expanded game without pretending private information is observable.

Example

Nature selects attacker type **Advanced** with probability **0.2** or **Basic** with probability **0.8**. The attacker observes its type; the defender does not. Both then act.

Common interview angle

Nature is not strategic and has no payoff. Its probability distribution encodes beliefs.

3.7 Common prior and correlated types

What it means

A **common prior** means all players' beliefs can be derived by conditioning one shared joint distribution. Types need not be independent.

Why it matters

With correlation, learning one's own type changes beliefs about other players. Replacing a joint prior with independent marginals can change the equilibrium.

Example

Two services share an upstream traffic shock. If service A observes high local demand, it should raise its belief that service B also has high demand.

Common interview angle

Common prior does not mean identical posteriors. Players can receive different information and therefore condition the same prior differently.

3.8 Interim and ex ante perspectives

What it means

- **Ex ante:** before anyone knows its type.
- **Interim:** after a player knows its own type but not others' types.
- **Ex post:** after all types are known.

Why it matters

BNE optimality is normally interim: every realized type optimizes given its conditional belief. Mechanism performance may be evaluated ex ante, while ex-post incentive compatibility is stronger.

Example

Before learning whether a job is urgent, a tenant assesses expected allocation quality ex ante. Once it learns urgency, it chooses a bid using interim expected payoff.

Common interview angle

Do not confuse high ex-ante average welfare with every type having no profitable interim deviation.

4. Real-World Example

First-price auction for cloud capacity

A cloud platform sells one immediate compute slot. Two jobs submit sealed bids. Each job privately knows its value v_i ; values are independently uniform on $[0, 100]$. The highest bid wins and pays its own bid.

The components are:

Component	Model
Players	Job 1 and Job 2
Type	Private value v_i
Prior	Each value is uniform on $[0, 100]$
Action	Bid $b_i \geq 0$
Winner payoff	$v_i - b_i$
Loser payoff	0
Strategy	A bid function $b_i(v_i)$

For two risk-neutral symmetric bidders, consider $b(v)=v/2$. If a bidder with value v pretends to be type x and bids $x/2$, it wins when the opponent value is below x , which has probability $x/100$. Expected payoff is:

$$\begin{aligned} EU(x; v) &= (v - x/2)(x/100) \\ &= (vx - x^2/2)/100 \end{aligned}$$

Differentiate with respect to x :

$$dEU/dx = (v - x)/100$$

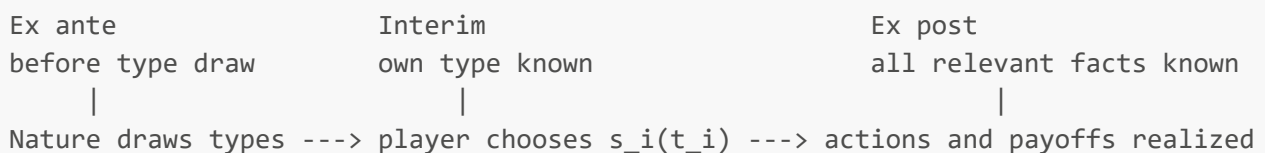
The maximum occurs at $x=v$, so bidding $v/2$ is a best response when the opponent uses the same rule. Therefore the symmetric strategy profile is a BNE.

Practical lessons:

- The action is a single bid, but the strategy is a bid for every possible value.
- Bidders shade bids below value because paying less is valuable.
- A higher bid improves win probability but reduces surplus.
- The result depends on the prior, payment rule, bidder count, independence, and risk attitude.

5. Diagrams / Mental Models

Information timeline



Strategy is a contingent plan

Player type	Strategy output
-----	-----
Low value	----> low bid
Medium value	----> medium bid
High value	----> high bid

The whole mapping is chosen as the equilibrium strategy.

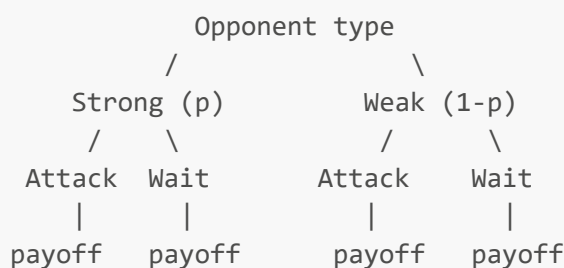
Solving checklist

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Identify players/actions
|
Identify private types and observations
|
Write joint prior and conditional beliefs
|
Propose type-contingent strategies
|
Compute each type's expected payoff from every deviation
|
All types best-respond? -- yes --> Bayesian Nash equilibrium
|
no
v
Revise candidate strategy

```

Probability tree



6. Common Interview Questions

Q1. What is incomplete information?

Answer: A game has incomplete information when some player lacks knowledge of a payoff-relevant characteristic, typically another player's type or payoff function. Types and beliefs model that uncertainty.

Interviewer expects: distinction between unknown types and merely unknown simultaneous actions.

Common mistake: saying every simultaneous game has incomplete information. Players can know the full payoff matrix while not knowing the action chosen at the same time.

Q2. What is a player type?

Answer: A type is a formal description of a player's privately known, payoff-relevant characteristics. Once the complete type profile is fixed, the model specifies payoffs and feasible decisions.

Interviewer expects: examples such as valuation, cost, capability, or risk preference.

Common mistake: treating a type as an action or as a vague personality label.

Q3. What are the components of a Bayesian game?

Answer: Players, action sets, type sets, a probability distribution over type profiles, information about what each player observes, and payoff functions depending on actions and types.

Interviewer expects: notation such as (N, A, T, p, u) and the role of Nature.

Common mistake: omitting the prior or information structure.

Q4. What is a prior probability?

Answer: It is the belief over possible types before observing additional evidence. In a common-prior model, each player's conditional beliefs derive from one joint distribution.

Interviewer expects: awareness that priors can be joint and correlated.

Common mistake: assuming independence without justification.

Q5. How is a posterior computed?

Answer: Use Bayes' rule: multiply the likelihood of the observation under a type by the prior probability of that type, then normalize across all types that could produce the observation.

Interviewer expects: $P(t|x) = P(x|t)P(t)/P(x)$.

Common mistake: reversing $P(x|t)$ and $P(t|x)$.

Q6. What is expected payoff?

Answer: It is the sum of each possible payoff multiplied by the probability of the state or action producing it. In Bayesian games, a type uses conditional beliefs over opponents' types.

Interviewer expects: a correct probability-weighted formula.

Common mistake: taking an unweighted arithmetic mean.

Q7. Define Bayesian Nash equilibrium.

Answer: It is a profile of type-contingent strategies such that, for every player and every type with positive probability, the prescribed action maximizes expected payoff given beliefs and the opponents' strategies.

Interviewer expects: "every type," "expected payoff," and "unilateral deviation."

Common mistake: checking one average player instead of each type.

Q8. How does BNE differ from ordinary Nash equilibrium?

Answer: Ordinary Nash equilibrium assumes the game's relevant payoff structure is known. BNE allows private types; strategies map types to actions and optimality is evaluated using beliefs. A Bayesian game can also be converted into an expanded complete-information strategic form over type-contingent plans.

Interviewer expects: the change in strategy space and expectation over hidden types.

Common mistake: claiming Nash equilibrium contains no uncertainty at all; mixed strategies can still introduce action uncertainty.

Q9. What is the Harsanyi transformation?

Answer: It models incomplete information as an initial chance move in which Nature draws types according to a prior, followed by players observing their permitted information and acting.

Interviewer expects: Nature is non-strategic and types make payoffs well-defined.

Common mistake: assigning Nature an objective.

Q10. Why is a Bayesian strategy a function rather than one action?

Answer: The player learns its type before acting, so the complete plan must specify an action for every type it might be. Equilibrium compares these full contingent plans.

Interviewer expects: $s_i: T_i \rightarrow A_i$.

Common mistake: listing only the action of the realized type.

Q11. Can players have different posterior beliefs under a common prior?

Answer: Yes. They may receive different private signals or observe different types. Conditioning the same joint prior on different information produces different posteriors.

Interviewer expects: common prior is not the same as common information.

Common mistake: assuming shared prior implies identical beliefs at every point.

Q12. How do you verify a candidate BNE?

Answer: Fix the opponents' candidate strategies. For each player and each possible type, compute conditional expected payoff for the prescribed action and every feasible deviation. The profile is a BNE only if none gives a strictly higher payoff.

Interviewer expects: type-by-type incentive checks.

Common mistake: verifying only outcomes that occur on the proposed path.

7. Deep-Dive Questions

Q1. What changes when types are correlated?

Observing t_i changes player i 's belief about t_{-i} , so the correct weight is $P(t_{-i} \mid t_i)$, not $P(t_{-i})$. Correlation can alter best responses even if each player's marginal type distribution stays unchanged.

Q2. Does every finite Bayesian game have a BNE?

Yes, if mixed strategies are allowed and players have finite type and action sets. Transform the Bayesian game into an agent-normal-form or strategic-form game whose pure strategies are type-to-action mappings; Nash's existence result then supplies a mixed equilibrium. A pure BNE need not exist.

Q3. What is the difference between BNE and sequential equilibrium?

BNE requires optimal type-contingent strategies given beliefs at the strategic-form level. Sequential equilibrium additionally requires sequential rationality at every information set and beliefs consistent with strategies, including careful treatment of off-path events. In dynamic games, BNE can permit non-credible future actions.

Q4. Why can actions reveal private types?

If types have different costs and benefits, they may prefer different actions. Observers infer type from action using Bayes' rule. A **separating** equilibrium uses different actions by type; a **pooling** equilibrium uses the same action and reveals less.

Q5. What if the prior assigns zero probability to a type or observation?

A zero-probability type does not affect expected payoff under that prior. Bayes' rule cannot determine beliefs after a zero-probability observation because the denominator is zero. Refinements for dynamic games impose additional consistency requirements for such off-path beliefs.

8. Comparison Tables

Complete versus incomplete information

Aspect	Complete information	Incomplete information
Payoff-relevant facts	Commonly known	Some are private or unknown
Basic private variable	None required	Type
Strategy	Action/contingent plan	Type-contingent plan
Optimization	Payoff against strategies	Expected payoff using beliefs
Equilibrium	Nash equilibrium	Bayesian Nash equilibrium
Example	Known payoff matrix	Auction with private values

Imperfect versus incomplete information

Aspect	Imperfect information	Incomplete information
Unknown item	Earlier actions/history	Types/payoff-relevant characteristics
Typical example	Player cannot observe a previous move	Bidder does not know rival's valuation
Representation	Information sets in extensive form	Types and beliefs
Can coexist?	Yes	Yes

Prior versus posterior

Aspect	Prior	Posterior
Timing	Before new evidence	After evidence

Aspect	Prior	Posterior
Formula role	Starting distribution	Updated conditional distribution
Example	20% of nodes are faulty	67% fault chance after timeout
Common trap	Assuming independence	Forgetting normalization

Nash equilibrium versus Bayesian Nash equilibrium

Aspect	Nash equilibrium	Bayesian Nash equilibrium
Private types	Not essential	Central
Strategy object	Strategy/action plan	Mapping from type to action
Payoff comparison	Usually direct	Conditional expected utility
Incentive check	Every player	Every type of every player
Beliefs	Usually implicit	Explicit prior/posterior

Ex ante, interim, and ex post

View	Information available	Typical question
Ex ante	No private type realized/known	Is the mechanism good on average?
Interim	Own type known	What action should this type choose?
Ex post	All types known	Would the action remain optimal after full revelation?

9. Common Mistakes

1. **Confusing hidden action with hidden type.** Simultaneous action uncertainty alone does not make the payoff structure incomplete.
2. **Treating a strategy as one action.** A Bayesian pure strategy specifies an action for every possible own type.
3. **Using unconditional probabilities.** With correlated types, use $P(t_{-i} \mid t_i)$.
4. **Averaging payoffs without weights.** Expected payoff must use the actual probability distribution.
5. **Checking only one type.** BNE requires a best response for every type with positive probability.
6. **Ignoring how an action changes beliefs.** In signaling settings, observed behavior conveys information.
7. **Assuming common prior means identical posterior.** Different information produces different conditioning.
8. **Using ordinal rankings as cardinal utilities.** Expected-utility arithmetic needs meaningful utility differences.
9. **Forgetting tie rules.** Auction equilibrium can depend on how equal bids are resolved.
10. **Assuming a pure BNE must exist.** Finite games guarantee equilibrium only when mixing is allowed.

10. Edge Cases / Special Cases

- **Degenerate prior:** If one type has probability 1, the model effectively reduces to complete information about that type.
- **Zero-probability types:** They do not constrain standard on-support BNE calculations, but may matter in refinements or robustness analysis.
- **Correlated types:** Own type is evidence about others; never multiply marginals automatically.
- **Continuous types:** Replace sums with integrals and often solve for threshold or monotone strategy functions.
- **Mixed strategies by type:** Each type may randomize differently, so the strategy maps a type to a probability distribution over actions.
- **Risk aversion:** Maximizing expected money is not the same as maximizing expected utility of money.
- **Interdependent values:** A player's payoff may depend on others' private information, not only its own value.
- **No common prior:** Players may hold fundamentally incompatible beliefs; standard common-prior Bayesian analysis may not apply directly.
- **Signals and noisy observations:** A posterior must condition on the signal-generation process, not merely on observed behavior.
- **Dynamic games:** BNE may not rule out non-credible continuation behavior; perfect Bayesian or sequential equilibrium may be needed.

11. How to Explain in Interview

"An incomplete-information game is one where players do not know all payoff-relevant facts, such as another bidder's valuation. We model each private fact as a type and place a prior over type profiles. A strategy maps each possible own type to an action. Each type chooses the action that maximizes conditional expected payoff given beliefs and the other players' type-contingent strategies. If every type is doing so, the strategy profile is a Bayesian Nash equilibrium."

12. Quick Revision Notes

Key definitions

- **Type:** private payoff-relevant information about a player.
- **Prior:** belief over type profiles before new evidence.
- **Posterior:** belief updated after information using Bayes' rule.
- **Bayesian strategy:** mapping $s_i: T_i \rightarrow A_i$ or to distributions over actions.
- **Expected payoff:** probability-weighted utility.
- **BNE:** every type's strategy is an expected-payoff best response.

Important points

- Nature draws types but does not optimize.
- Conditional beliefs matter when types are correlated.
- Verify deviations one player and one type at a time.
- Strategies, not realized actions alone, form the equilibrium.
- Pure BNE may fail to exist; mixed BNE exists in finite games.

Common comparisons

- Complete information concerns knowledge of payoffs/types; perfect information concerns observation of history.
- Prior is before evidence; posterior is after conditioning.
- Nash uses known game structure; BNE explicitly includes private types and beliefs.

Must-remember facts

$$EU_i(a_i | t_i) = \sum P(t_{-i} | t_i) u_i(a_i, s_{-i}(t_{-i}), t_i)$$

BNE = no profitable unilateral deviation for any player type

Interview traps

- Do not say “probability of an action” when the question asks for a prior over types.
- Do not forget to normalize a posterior.
- Do not substitute expected type into a nonlinear payoff.
- Do not verify only the most likely type.

13. Practice Tasks

1. **Type-to-strategy enumeration:** Two types and two actions give four pure strategies. List them explicitly.
2. **Posterior calculation:** Start with a 10% malicious-node prior and two signal likelihoods; compute the posterior after an alert.
3. **Expected payoff table:** For two defender actions and two attacker types, calculate each action's expected payoff as a function of prior p .
4. **Threshold derivation:** Find the value of p at which the defender switches from monitoring to blocking.
5. **BNE verification:** Given candidate strategies for high- and low-cost sellers, check every type's deviations.
6. **Auction derivation:** Re-derive $b(v)=v/2$ for two uniform first-price bidders.
7. **Simulation:** Write a short C++ or Python program that samples bidder values, applies $b(v)=v/2$, and estimates average utility.
8. **Correlation exercise:** Construct a joint type table where seeing your own high demand raises the chance the other service has high demand.
9. **Information comparison:** Convert a complete-information matrix into a Bayesian game by making one payoff parameter private.
10. **Interview drill:** Explain why $s_i(t_i)$ is a function in under 30 seconds.

14. Final Cheat Sheet

Item	Remember
Core definition	Incomplete information means some payoff-relevant fact is unknown to at least one player.
Type	Formal package of a player's private information.
Bayesian game	Players + actions + types + prior + information + payoffs.
Prior	Distribution over type profiles before evidence.
Expected payoff	Sum/integral of payoff weighted by conditional probabilities.
BNE	Every type of every player best-responds in expected utility.
Main comparison	Nash: known payoff structure; BNE: private types and beliefs.
Most asked question	How do you verify BNE? Check every type's expected payoff against every unilateral deviation.
Biggest trap	Treating a Bayesian strategy as one realized action.
One-line answer	"BNE is Nash equilibrium adapted to private types: each type chooses an expected-payoff best response given beliefs."